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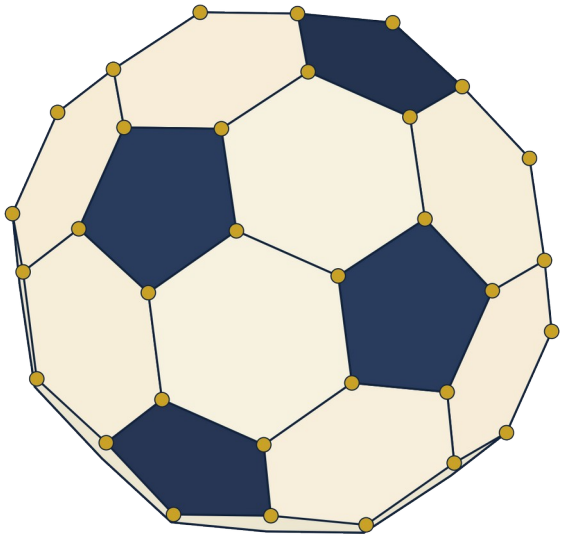
S.K. Nagar, Motihari

CLASS 10 • SCIENCE (CHEMISTRY)

Chapter 4

CARBON AND ITS COMPOUNDS

Complete Chapter Notes • CBSE Session 2026–27



Buckminsterfullerene (C_{60}) — sixty carbon atoms arranged like a football

INSIDE THESE NOTES

- Covalent bonding & electron dot structures
- Versatile nature: catenation & tetravalency
- Functional groups & homologous series
- Combustion, oxidation, addition, substitution
- Soaps, detergents & micelles
- Allotropes: diamond, graphite, fullerene
- Hydrocarbons, chains, rings & isomers
- IUPAC nomenclature with worked examples
- Ethanol & ethanoic acid
- Mind map, NCERT solutions & practice set

Name: _____ **Batch:** _____

CBSE SYLLABUS 2026–27 **Unit I — Chemical Substances: Nature and Behaviour**

Unit I carries **25 of the 80 marks** in the theory paper (shared with Chapters 1–3). Topics prescribed for this chapter:

- Covalent bonds — formation and properties of covalent compounds; versatile nature of carbon
- Hydrocarbons — saturated and unsaturated; homologous series
- Nomenclature of alkanes, alkenes, alkynes and compounds containing functional groups (halogens, alcohols, aldehydes, ketones)
- Difference between saturated and unsaturated hydrocarbons
- Chemical properties — combustion, oxidation, addition and substitution reactions
- Ethanol and ethanoic acid (only properties and uses); soaps and detergents

Contents

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8	Homologous Series	17	Practice Zone with Answer Key
9	Nomenclature of Carbon Compounds		

How to use these notes

Key terms are printed in **bold**. Coloured boxes mark special points:

REMEMBER	EXAM TIP	COMMON MISTAKE	ACTIVITY	DO YOU KNOW?
Must-know facts	Scoring hints for the board exam	Errors students often make	NCERT activity with result	Extra facts for curiosity

EXAM TIP **Study plan for this chapter**

1. Learn the electron dot structures (Section 2) — they are frequently asked in board exams.
2. Master the naming rules with the propane family table (Section 9) and practise both directions: name → structure and structure → name.
3. Write each reaction of Sections 10–12 three times with its condition (catalyst, sunlight, heat, conc. H_2SO_4).
4. Revise with the Mind Map and Reaction Bank (Section 15) the night before the test.

1 CARBON — AN INTRODUCTION

Carbon is one of the most important elements for life. Every living thing — plants, animals and our own bodies — is built from **carbon compounds**. Look around you: almost everything you have used or eaten since morning contains carbon.

Where we meet carbon	Examples
Food	Carbohydrates (rice, wheat, sugar), proteins, fats and oils, vitamins
Clothes	Cotton, wool, silk, nylon, polyester
Fuels	Wood, coal, LPG, CNG, biogas, petrol, diesel, kerosene
Other materials	Paper, plastics, rubber, medicines, soaps, detergents, paints

1.1 Occurrence of carbon

- **Free (elemental) state:** diamond, graphite and fullerenes. Coal and charcoal contain impure carbon.
- **Combined state:** carbon dioxide in air; carbonates such as limestone and marble ($CaCO_3$) and hydrogencarbonates; fossil fuels (coal, petroleum, natural gas); and all living things.
- The earth's crust has only **0.02% carbon** (in minerals like carbonates, hydrogencarbonates, coal and petroleum) and the atmosphere has only **0.03% carbon dioxide**.
- In spite of this small amount, carbon is immensely important. The special properties that explain this are studied in this chapter.

1.2 The carbon atom

Symbol	Atomic number	Atomic mass	Electronic configuration	Valence electrons	Valency
C	6	12 u	K = 2, L = 4	4	4 (tetravalent)

Carbon is a **non-metal**. With 4 electrons in its outermost shell, it needs 4 more electrons to complete its octet — this decides the type of bonds it forms (Section 2).

1.3 Organic compounds

Compounds of carbon are called **organic compounds** and their study is called **organic chemistry**. Earlier, scientists believed that organic compounds could be made only inside living bodies by a mysterious **"vital force"** (vital force theory).

- In **1828, Friedrich Wöhler** prepared **urea** (an organic compound found in urine) in the laboratory by heating **ammonium cyanate** (an inorganic compound). This disproved the vital force theory.

$$NH_4CNO \xrightarrow{\text{heat}} NH_2CONH_2$$

Ammonium cyanate → Urea

> **REMEMBER**
>
> Carbides, oxides of carbon (CO, CO_2), carbonates and hydrogencarbonates contain carbon but are studied as **inorganic** compounds. All other carbon compounds are studied under organic chemistry.

2 BONDING IN CARBON: THE COVALENT BOND

In Chapter 3 we saw that ionic compounds have high melting and boiling points and conduct electricity in solution or in the molten state. Most carbon compounds behave very differently:

Carbon compound	Melting point (K)	Boiling point (K)
Acetic acid (CH_3COOH)	290	391
Chloroform ($CHCl_3$)	209	334
Ethanol (CH_3CH_2OH)	156	351
Methane (CH_4)	90	111

- **Low melting and boiling points** → the forces of attraction between their molecules are weak.
- **Poor conductors of electricity** → the bonding in these compounds does not give rise to ions.

2.1 Why can't carbon form ionic bonds?

Carbon (2, 4) has **4 valence electrons**. To attain the stable noble-gas configuration (octet), it would have to gain or lose 4 electrons — both are very difficult:

(i) Gain 4 electrons → C^{4-} anion	(ii) Lose 4 electrons → C^{4+} cation
It would be difficult for a nucleus with only **6 protons** to hold on to **10 electrons** (4 extra electrons).	Removing 4 electrons needs a **very large amount of energy**; the C^{4+} ion would have 6 protons holding on to just 2 electrons.

So carbon **shares** its valence electrons with other atoms of carbon or with atoms of other elements. The shared electrons belong to the outer shells of **both** atoms, so both attain the noble-gas configuration. Many other elements (H, O, N, Cl …) also form molecules by sharing electrons.

> DEFINITION
>
> **Covalent bond:** A chemical bond formed by the **sharing of a pair of electrons** between two atoms. Compounds formed by covalent bonds are called **covalent compounds**.

2.2 Single, double and triple bonds

Type of bond	Electrons shared	Shown as	Examples
Single bond	1 pair (2 electrons)	one line (–)	H–H, Cl–Cl, C–H bonds in CH_4
Double bond	2 pairs (4 electrons)	two lines (=)	O=O, C=C in ethene, O=C=O
Triple bond	3 pairs (6 electrons)	three lines (≡)	N≡N, C≡C in ethyne

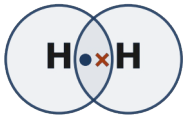
> REMEMBER **Electron dot (Lewis) structures**
>
> - Show **only the valence (outermost-shell) electrons** as dots (•) or crosses (×). Different symbols help us see which atom contributed which electron.
> - A **shared pair** is drawn between the two atoms; unshared pairs are called **lone pairs**.
> - After bonding, every atom should have **8 electrons** around it (hydrogen: **2 electrons**).

2.3 Valency from electronic configuration

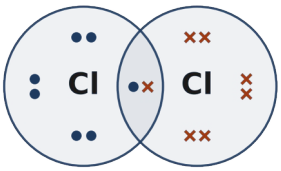
Element	Atomic no.	Configuration	Valence electrons	Electrons needed	Valency
Hydrogen (H)	1	1	1	1 (duplet, like He)	1
Carbon (C)	6	2, 4	4	4	4
Nitrogen (N)	7	2, 5	5	3	3
Oxygen (O)	8	2, 6	6	2	2
Fluorine (F)	9	2, 7	7	1	1
Sulphur (S)	16	2, 8, 6	6	2	2
Chlorine (Cl)	17	2, 8, 7	7	1	1

2.4 Formation of some covalent molecules

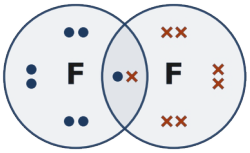
1. **Hydrogen (H_2):** H (atomic no. 1) has 1 electron and needs 1 more to fill its K shell (like helium). Two H atoms share one electron each → one shared pair → **single bond**, H–H.
2. **Chlorine (Cl_2):** Cl (2, 8, 7) needs 1 electron. Each Cl atom shares 1 electron → **single bond**, Cl–Cl. Each Cl keeps 3 lone pairs. **Fluorine (F_2)** is formed in the same way.
3. **Oxygen (O_2):** O (2, 6) needs 2 electrons. Each O atom shares 2 electrons → 2 shared pairs → **double bond**, O=O.
4. **Nitrogen (N_2):** N (2, 5) needs 3 electrons. Each N atom shares 3 electrons → 3 shared pairs → **triple bond**, N≡N.
5. **Water (H_2O):** O shares one electron with each of the two H atoms → two O–H single bonds; O keeps 2 lone pairs. **H_2S** is formed in the same way.
6. **Ammonia (NH_3):** N shares one electron with each of three H atoms → three N–H single bonds; N keeps 1 lone pair.
7. **Methane (CH_4):** C shares its 4 valence electrons with 4 H atoms → four C–H single bonds. Methane is a major part of **biogas** and **CNG** and is widely used as a fuel.
8. **Carbon dioxide (CO_2):** C shares 2 electrons with each O atom → two double bonds, O=C=O.
9. **Chloromethane (CH_3Cl):** C shares 1 electron with each of 3 H atoms and 1 electron with Cl → four single covalent bonds.



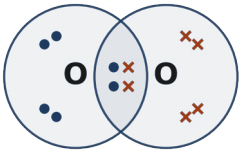
H_2 — single bond (H–H)



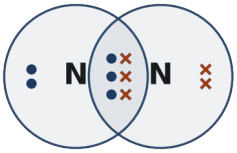
Cl_2 — single bond (Cl–Cl)



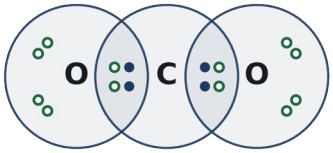
F_2 — single bond (F–F)



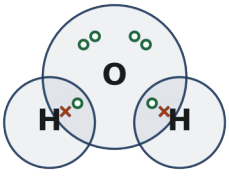
O_2 — double bond (O=O)



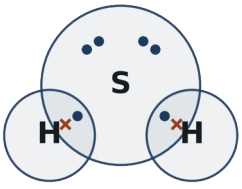
N_2 — triple bond (N≡N)



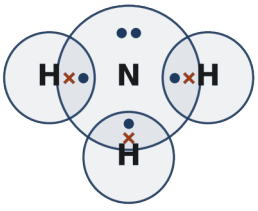
CO_2 — two double bonds



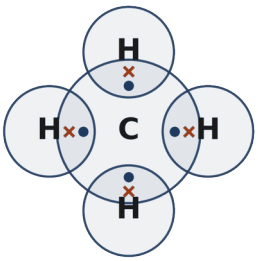
H_2O — two single bonds



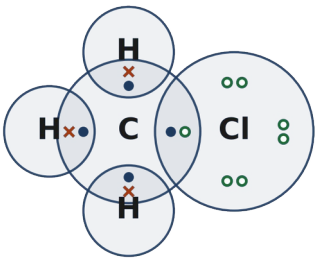
H_2S — two single bonds



NH_3 — three single bonds



CH_4 — four single bonds



CH_3Cl — four single bonds

Key: • × o = valence electrons of different atoms. The shaded overlap of two circles holds the shared pair(s). Lone pairs stay inside each atom's own circle.

2.5 Properties of covalent compounds

1. **Low melting and boiling points:** the bonds within a molecule are strong, but the forces between molecules (intermolecular forces) are weak.
2. **Poor conductors of electricity:** electrons are shared and no charged particles (ions) are formed.
3. **Solubility:** generally insoluble (or less soluble) in water but soluble in organic solvents such as kerosene and benzene. (Ethanol, ethanoic acid and sugar are exceptions that dissolve in water.)
4. **Physical state:** usually gases, liquids or soft solids at room temperature.

Ionic compounds	Covalent compounds
Formed by **transfer** of electrons	Formed by **sharing** of electrons
Made of ions (e.g. Na^+ and Cl^-)	Made of molecules
High melting and boiling points	Low melting and boiling points
Conduct electricity in molten state or in solution	Generally do not conduct electricity
Usually soluble in water, insoluble in kerosene/petrol	Usually insoluble in water, soluble in organic solvents
Hard, crystalline solids	Gases, liquids or soft solids
Examples: NaCl, MgO, $CaCl_2$	Examples: CH_4, CO_2, C_2H_5OH

EXAM TIP **Counting covalent bonds quickly**

- In an **alkane C_nH_{2n+2}**: number of covalent bonds = **3n + 1** → ethane (n = 2): **7**; propane: 10; butane: 13; pentane: 16.
- Counting **shared electron pairs** (a double bond = 2 pairs, a triple bond = 3 pairs): alkene C_nH_{2n} → **3n**; alkyne C_nH_{2n-2} → **3n − 1**. Example: C_2H_4 has C=C (2 pairs) + 4 C–H = 6 pairs.

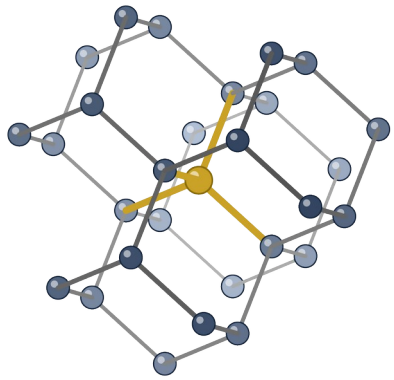
3 ALLOTROPES OF CARBON

DEFINITION

Allotropy: The property by which an element exists in **two or more forms** that have **different physical properties** but **similar chemical properties**. These different forms are called **allotropes**.

Carbon occurs in different forms in nature with widely varying physical properties. Diamond, graphite and fullerenes are all made of carbon atoms — the difference lies only in **the way the carbon atoms are bonded to one another**.

3.1 Diamond

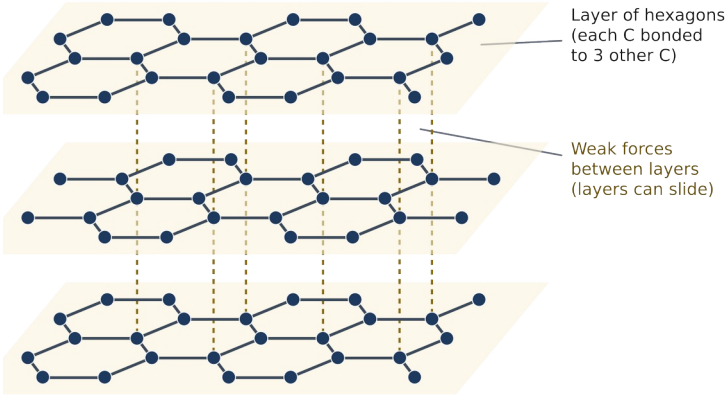


Diamond: the gold atom is bonded to four others

- Each carbon atom is bonded to **four other carbon atoms** (single covalent bonds), forming a **rigid three-dimensional network** (tetrahedral arrangement).
- **Hardest natural substance** known; very high melting point.
- **Bad conductor of electricity** — all 4 valence electrons are used in bonding; no free electrons.
- Transparent; sparkles brilliantly when cut and polished.
- **Uses:** jewellery; cutting glass and drilling rocks (diamond-tipped tools).
- **Synthetic diamonds** are made by subjecting pure carbon to very high pressure and temperature. They are small but otherwise indistinguishable from natural diamonds.

3.2 Graphite

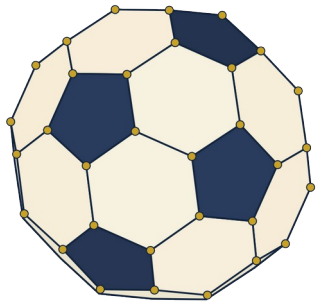
- Each carbon atom is bonded to **three other carbon atoms in the same plane**, giving a **hexagonal array**. One of these bonds is a **double bond**, so the valency of carbon is satisfied.
- Hexagonal layers lie one above another, held by **weak forces** → layers slide → **soft and slippery**.
- **Good conductor of electricity** — unlike other non-metals (it has free, mobile electrons).



Graphite: layers of hexagons

- Black, opaque and smooth/greasy to touch.
- **Uses:** dry lubricant for machine parts; electrodes in dry cells and electrolysis; the “lead” of pencils (graphite mixed with clay).

3.3 Fullerenes



C_{60} fullerene

- Fullerenes form another class of carbon allotropes. The first one identified was **C_{60} (buckminsterfullerene)**, in which **60 carbon atoms** are arranged in the shape of a **football** (12 pentagons and 20 hexagons).
- Its structure looked like the **geodesic dome** designed by the US architect **Buckminster Fuller**, so the molecule was named **fullerene**.
- Each carbon atom is bonded to three other carbon atoms.

3.4 Diamond vs graphite

Property	Diamond	Graphite
Bonding of each C atom	Bonded to **4** other C atoms	Bonded to **3** other C atoms
Structure	Rigid 3-D network	Flat hexagonal layers held by weak forces
Hardness	Hardest natural substance	Soft and slippery
Electrical conductivity	Non-conductor	Good conductor
Appearance	Transparent, brilliant	Black, opaque, shiny
Uses	Jewellery, cutting and drilling tools	Lubricant, electrodes, pencil lead

REMEMBER

All allotropes of carbon have the **same chemical properties**. Each burns in oxygen to give only carbon dioxide — proof that they are all made of carbon:

$$C + O_2 \longrightarrow CO_2 + \text{heat and light}$$

DO YOU KNOW?

Coal, charcoal, coke and lamp black (soot) are **amorphous** (non-crystalline) forms of carbon. **Graphene** is a single layer of graphite only one atom thick; it is extremely strong and an excellent conductor.

4 VERSATILE NATURE OF CARBON

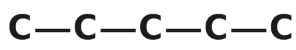
According to NCERT, the number of carbon compounds whose formulae are known to chemists is about **three million** — far more than the compounds formed by all the other elements put together! The nature of the covalent bond allows carbon to form so many compounds. Two features are responsible:

4.1 Catenation

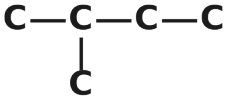
DEFINITION

Catenation: The unique ability of carbon to form covalent bonds with **other carbon atoms**, giving rise to large molecules — **long chains, branched chains or rings**.

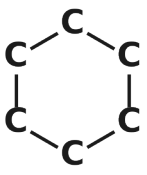
- Carbon atoms may be linked by **single, double or triple bonds**.
- The C–C bond is **very strong and stable**, so long chains of carbon atoms are possible.
- No other element shows catenation to this extent. **Silicon** forms compounds with hydrogen having chains of only **7 or 8 atoms**, and these compounds are very reactive.
- Compounds with only single C–C bonds are **saturated**; compounds with double or triple C–C bonds are **unsaturated**.



Straight (open) chain



Branched chain



Ring (closed chain)

4.2 Tetravalency

- Carbon has a valency of **4**, so it can bond with **four other atoms** — of carbon or of monovalent elements such as hydrogen and chlorine.

- Carbon forms compounds with **oxygen, hydrogen, nitrogen, sulphur, chlorine** and many other elements. The specific properties of each compound depend on the elements other than carbon present in it.
- The bonds formed by carbon with most other elements are **very strong**, which makes these compounds exceptionally stable.

4.3 Why are carbon's bonds so strong?

One reason is the **small size** of the carbon atom. Its nucleus holds the shared pairs of electrons strongly. Bonds formed by elements with bigger atoms are much weaker.

REMEMBER

Catenation + tetravalency (together with strong bonds due to small size) → an enormous number of **stable** carbon compounds.

EXAM TIP

Q: "What are the two properties of carbon which lead to the huge number of carbon compounds?" → **Catenation and tetravalency.** Write one line explaining each for full marks.

5 HYDROCARBONS: SATURATED AND UNSATURATED

DEFINITION

Hydrocarbons: Compounds made up of **only carbon and hydrogen**. Their main sources are petroleum and natural gas.

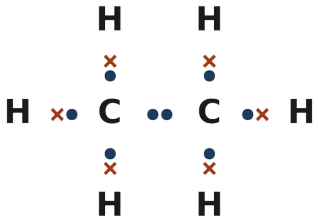
	Alkanes	Alkenes	Alkynes
C–C bonds	Only single bonds	One double bond (C=C)	One triple bond (C≡C)
Nature	Saturated	Unsaturated	Unsaturated
General formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}
Name ends in	-ane	-ene	-yne
First member	Methane, CH_4	Ethene, C_2H_4	Ethyne, C_2H_2
Reactivity	Less reactive	More reactive	More reactive

5.1 Saturated hydrocarbons — alkanes

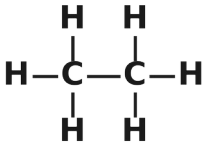
Carbon compounds in which the carbon atoms are linked by **only single bonds** are **saturated**. All the valencies of the atoms are satisfied by single bonds, so these compounds are normally **not very reactive**.

EXAM TIP **How to draw the structure of a carbon compound (NCERT method)**

1. Link the carbon atoms together with **single bonds** (the carbon "skeleton").
2. Use **hydrogen atoms** to satisfy the remaining valencies of each carbon (4 bonds per C).
3. If there are not enough H atoms, put a **double or triple bond** between carbon atoms so that every carbon still has 4 bonds.

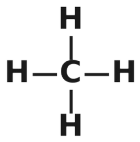


Electron dot structure of **ethane** (C_2H_6): 7 shared pairs

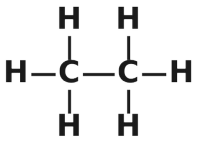


Structural formula of ethane

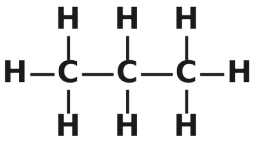
n	Name	Molecular formula	Condensed formula	Mol. mass (u)	State (room temp.)
1	Methane	CH_4	CH_4	16	Gas
2	Ethane	C_2H_6	$CH_3–CH_3$	30	Gas
3	Propane	C_3H_8	$CH_3–CH_2–CH_3$	44	Gas
4	Butane	C_4H_{10}	$CH_3–CH_2–CH_2–CH_3$	58	Gas
5	Pentane	C_5H_{12}	$CH_3–(CH_2)_3–CH_3$	72	Liquid
6	Hexane	C_6H_{14}	$CH_3–(CH_2)_4–CH_3$	86	Liquid
7	Heptane	C_7H_{16}	$CH_3–(CH_2)_5–CH_3$	100	Liquid
8	Octane	C_8H_{18}	$CH_3–(CH_2)_6–CH_3$	114	Liquid
9	Nonane	C_9H_{20}	$CH_3–(CH_2)_7–CH_3$	128	Liquid
10	Decane	$C_{10}H_{22}$	$CH_3–(CH_2)_8–CH_3$	142	Liquid



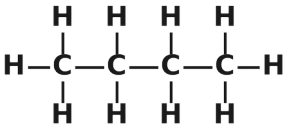
Methane, CH_4



Ethane, C_2H_6



Propane, C_3H_8

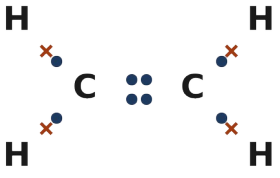


Butane, C_4H_{10}

5.2 Unsaturated hydrocarbons — alkenes and alkynes

Ethene (C_2H_4): if we join the two carbon atoms by a single bond and attach the four hydrogen atoms, each carbon still has one unsatisfied valency. This is satisfied only if there is a **double bond** between the two carbon atoms. In the same way, **ethyne (C_2H_2)** needs a **triple bond**.

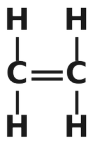
Compounds of carbon having **double or triple bonds** between carbon atoms are **unsaturated**. They are **more reactive** than saturated compounds.



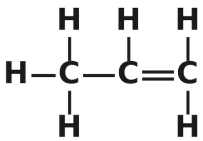
Electron dot structure of **ethene** (C=C)



Electron dot structure of **ethyne** (C≡C)

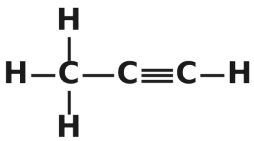


Ethene, C_2H_4



Propene, C_3H_6

Ethyne, C_2H_2



Propyne, C_3H_4

Alkene	Formula	Condensed formula	Alkyne	Formula	Condensed formula
Ethene	C_2H_4	$CH_2{=}CH_2$	Ethyne	C_2H_2	$HC{\equiv}CH$
Propene	C_3H_6	$CH_3{-}CH{=}CH_2$	Propyne	C_3H_4	$CH_3{-}C{\equiv}CH$
Butene	C_4H_8	$CH_3{-}CH_2{-}CH{=}CH_2$	Butyne	C_4H_6	$CH_3{-}CH_2{-}C{\equiv}CH$
Pentene	C_5H_{10}	$CH_3{-}(CH_2)_2{-}CH{=}CH_2$	Pentyne	C_5H_8	$CH_3{-}(CH_2)_2{-}C{\equiv}CH$

COMMON MISTAKE

There is **no "methene" or "methyne"**. A double or triple bond needs at least **two** carbon atoms, so the alkene and alkyne series start with **2 carbons**.

5.3 Saturated vs unsaturated hydrocarbons

Saturated hydrocarbons	Unsaturated hydrocarbons
Only **single** bonds between carbon atoms	At least one **double (C=C)** or **triple (C≡C)** bond
Alkanes: C_nH_{2n+2}	Alkenes: C_nH_{2n}; alkynes: C_nH_{2n-2}
Less reactive (fairly unreactive)	More reactive
Undergo **substitution** reactions	Undergo **addition** reactions
Burn with a **clean blue flame**	Burn with a **yellow, sooty (smoky) flame**
Do not decolourise bromine water	**Decolourise** reddish-brown bromine water
Examples: methane, ethane, propane	Examples: ethene, propene, ethyne

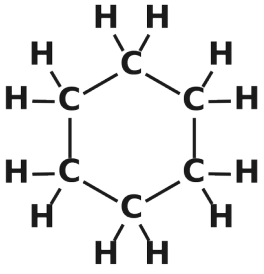
EXAM TIP **Test to distinguish saturated and unsaturated compounds**

- **Bromine water test:** shake the compound with a few drops of bromine water (reddish-brown). An unsaturated compound **decolourises** it (addition of bromine); a saturated compound does not.
- **Flame test:** burn the compound on a spatula — unsaturated compounds give a **sooty flame** and leave a black deposit on a plate held above.

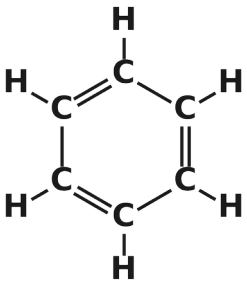
6 CHAINS, BRANCHES, RINGS AND ISOMERS

6.1 Chains, branches and rings

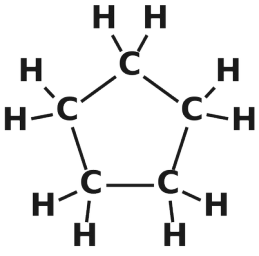
Carbon atoms can be arranged in **straight chains**, **branched chains** or **rings** (cyclic compounds). All three types may be **saturated or unsaturated**.



Cyclohexane, C_6H_{12} — saturated ring



Benzene, C_6H_6 — unsaturated ring (alternate double bonds)



Cyclopentane, C_5H_{10} — saturated ring

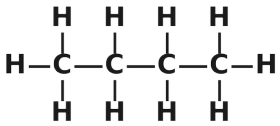
- Compounds containing only carbon and hydrogen are **hydrocarbons**: saturated ones are **alkanes**; those with one or more double bonds are **alkenes**; those with one or more triple bonds are **alkynes**.
- Ring compounds of alkanes (cycloalkanes) have the general formula **C_nH_{2n}** — the same as alkenes — even though they are saturated.

6.2 Structural isomerism

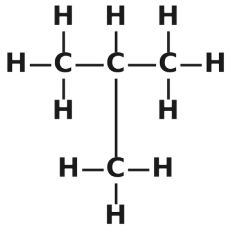
DEFINITION

Structural isomers: Compounds having the **same molecular formula** but **different structures** (different arrangement of atoms). The phenomenon is called **structural isomerism**.

Butane (C_4H_{10}) — with four carbon atoms, two different carbon skeletons are possible:

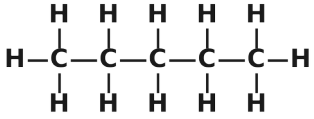


n-Butane (straight chain) — boils at about −0.5 °C

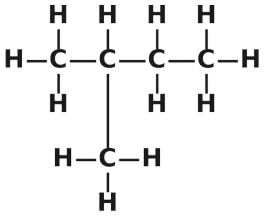


Isobutane (branched; 2-methylpropane) — boils at about −12 °C

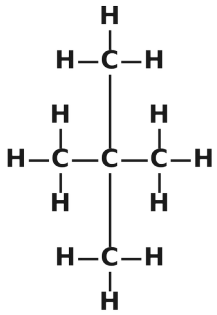
Pentane (C_5H_{12}) has **three** structural isomers:



n-Pentane (straight chain)



Isopentane (2-methylbutane)



Neopentane (2,2-dimethylpropane)

- Methane, ethane and propane have **no isomers** — isomerism in alkanes starts with **butane (C_4)**.
- The number of isomers rises quickly: C_4H_{10} → 2, C_5H_{12} → 3, C_6H_{14} → 5.
- Isomers have **different physical properties** (e.g. boiling points), as shown above.

COMMON MISTAKE

Bending a chain on paper does **not** make a new isomer! $CH_3–CH_2–CH_2–CH_3$ drawn with a bend is still n-butane. Check the **longest continuous carbon chain** and the **position of branches** before calling two structures isomers.

7 FUNCTIONAL GROUPS

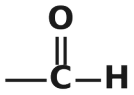
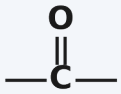
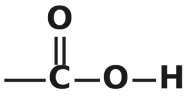
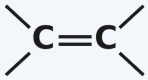
Carbon is a very "friendly" element. Besides hydrogen, it forms bonds with **halogens, oxygen, nitrogen and sulphur**. In a hydrocarbon chain, one or more hydrogen atoms can be replaced by these elements so that the valency of carbon remains satisfied.

DEFINITION

Heteroatom: An element other than carbon and hydrogen (e.g. Cl, Br, O, N, S) that replaces hydrogen in a hydrocarbon chain.

DEFINITION

Functional group: An atom or group of atoms (containing a heteroatom, or a C=C / C≡C bond) that gives a carbon compound its **characteristic properties, regardless of the length and nature of the carbon chain**.

Heteroatom	Class of compound	Functional group	Structure	Name ending	Example
Cl / Br	Haloalkane	–Cl, –Br (–X)	—X	prefix **chloro-, bromo-**	CH_3Cl chloromethane
Oxygen	Alcohol	–OH	—O—H	suffix **-ol**	C_2H_5OH ethanol
Oxygen	Aldehyde	–CHO		suffix **-al**	CH_3CHO ethanal
Oxygen	Ketone	–CO–		suffix **-one**	CH_3COCH_3 propanone
Oxygen	Carboxylic acid	–COOH		suffix **-oic acid**	CH_3COOH ethanoic acid
—	Alkene	C=C		suffix **-ene**	$CH_2=CH_2$ ethene
—	Alkyne	C≡C		suffix **-yne**	HC≡CH ethyne

The free valency of a group is shown by a single line. The group joins the carbon chain through this valency by replacing a hydrogen atom.

REMEMBER

- The **–CHO** (aldehyde) and **–COOH** (carboxylic acid) groups are always at the **end** of a chain, and their carbon atom is **counted** in the chain.
- The **ketone group (–CO–)** has two free valencies, so it is always **inside** the chain. The smallest ketone therefore has **3 carbon atoms** (propanone).
- The letter **R** stands for any alkyl group: R–OH (alcohol), R–CHO (aldehyde), R–CO–R (ketone), R–COOH (acid), R–X (haloalkane).

Common names you should know

IUPAC name	Common name	Where you meet it
Methanal (HCHO)	Formaldehyde	Its 40% solution, **formalin**, preserves biological specimens
Ethanal (CH_3CHO)	Acetaldehyde	Used to make other chemicals
Propanone (CH_3COCH_3)	Acetone	Solvent; nail-polish remover
Methanoic acid (HCOOH)	Formic acid	Present in **ant stings** and **nettle** leaves
Ethanoic acid (CH_3COOH)	Acetic acid	**Vinegar**
Ethanol (C_2H_5OH)	Ethyl alcohol	Medicines, fuel, alcoholic drinks
Ethene (C_2H_4) / Ethyne (C_2H_2)	Ethylene / Acetylene	Ripening fruit, plastics / welding

8 HOMOLOGOUS SERIES

DEFINITION

Homologous series: A series of carbon compounds in which the **same functional group** substitutes for hydrogen in a carbon chain. Successive members differ by a **$-CH_2-$ unit**.

Example — alcohols: CH_3OH, C_2H_5OH, C_3H_7OH and C_4H_9OH all contain the –OH group, and their chemical properties are very similar.

Pair of successive members	Difference in formula	Difference in molecular mass
CH_3OH (32 u) and C_2H_5OH (46 u)	$-CH_2-$	14 u
C_2H_5OH (46 u) and C_3H_7OH (60 u)	$-CH_2-$	14 u
C_3H_7OH (60 u) and C_4H_9OH (74 u)	$-CH_2-$	14 u

Atomic masses: C = 12 u, H = 1 u, O = 16 u. So one CH_2 unit = 12 + 2 = 14 u.

8.1 Characteristics of a homologous series

1. All members have the **same functional group**.
2. All members can be represented by the **same general formula**.
3. Successive members differ by a **$-CH_2-$ unit** in their molecular formula.
4. Successive members differ by **14 u** in molecular mass.
5. All members have **similar chemical properties** (decided by the functional group).
6. Physical properties show a **gradation** (gradual change): melting and boiling points **increase** with increasing molecular mass; solubility in a particular solvent also changes gradually.

8.2 General formulae

Homologous series	General formula	First member	Smallest n
Alkanes	C_nH_{2n+2}	CH_4 (methane)	1
Alkenes	C_nH_{2n}	C_2H_4 (ethene)	2
Alkynes	C_nH_{2n-2}	C_2H_2 (ethyne)	2
Haloalkanes	$C_nH_{2n+1}X$ (X = Cl, Br)	CH_3Cl (chloromethane)	1
Alcohols	$C_nH_{2n+1}OH$	CH_3OH (methanol)	1
Aldehydes	$C_nH_{2n}O$	HCHO (methanal)	1
Ketones	$C_nH_{2n}O$	CH_3COCH_3 (propanone)	3
Carboxylic acids	$C_nH_{2n}O_2$	HCOOH (methanoic acid)	1

Here n = total number of carbon atoms in the molecule.

8.3 Homologous series up to four carbon atoms

C atoms	Haloalkane	Alcohol	Aldehyde	Ketone	Carboxylic acid
1	CH_3Cl chloromethane	CH_3OH methanol	HCHO methanal	—	HCOOH methanoic acid
2	C_2H_5Cl chloroethane	C_2H_5OH ethanol	CH_3CHO ethanal	—	CH_3COOH ethanoic acid
3	C_3H_7Cl chloropropane	C_3H_7OH propanol	C_2H_5CHO propanal	CH_3COCH_3 propanone	C_2H_5COOH propanoic acid
4	C_4H_9Cl chlorobutane	C_4H_9OH butanol	C_3H_7CHO butanal	$CH_3COC_2H_5$ butanone	C_3H_7COOH butanoic acid

COMMON MISTAKE

Isomers are not homologues. Butane and isobutane (both C_4H_{10}) are **isomers**; butane (C_4H_{10}) and pentane (C_5H_{12}) are **homologues** — they differ by CH_2.

9 NOMENCLATURE OF CARBON COMPOUNDS

Carbon compounds are named by the **IUPAC** system (International Union of Pure and Applied Chemistry). The name of a compound in a homologous series is based on the name of the basic carbon chain, modified by a **prefix** (phrase before) or **suffix** (phrase after) that shows the functional group.

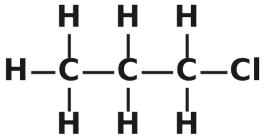
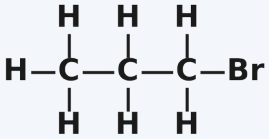
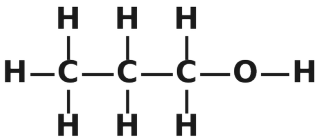
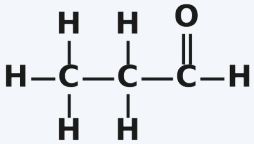
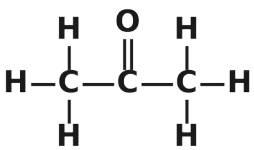
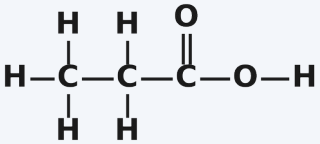
Word roots — number of carbon atoms

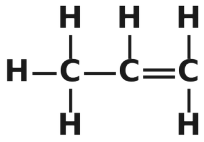
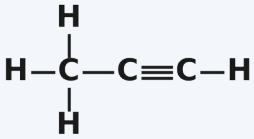
C atoms	1	2	3	4	5	6	7	8	9	10
Root	Meth	Eth	Prop	But	Pent	Hex	Hept	Oct	Non	Dec

Rules for naming (NCERT)

1. **Count the carbon atoms** in the chain (include the carbon of –CHO, –COOH or –CO–) and write the name of the parent alkane. Three carbons → **propane**.
2. Show the **functional group** by a prefix or a suffix (table below).
3. If the suffix begins with a **vowel** (a, e, i, o, u), **drop the final "e"** of the parent name: propane − e + one = **propanone**; propane − e + ol = **propanol**.
4. If the carbon chain is **unsaturated**, replace "ane" by **"ene"** (double bond) or **"yne"** (triple bond): **propene**, **propyne**.

Nomenclature table — the propane family

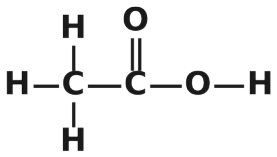
Class	Prefix / suffix	Formula	Name	Structure
Haloalkane	Prefix: chloro, bromo	$CH_3CH_2CH_2Cl$	**Chloropropane**	
		$CH_3CH_2CH_2Br$	**Bromopropane**	
Alcohol	Suffix: -ol	$CH_3CH_2CH_2OH$	**Propanol**	
Aldehyde	Suffix: -al	CH_3CH_2CHO	**Propanal**	
Ketone	Suffix: -one	CH_3COCH_3	**Propanone**	
Carboxylic acid	Suffix: -oic acid	CH_3CH_2COOH	**Propanoic acid**	

Class	Prefix / suffix	Formula	Name	Structure
Alkene	Suffix: -ene	$CH_3CH=CH_2$	**Propene**	
Alkyne	Suffix: -yne	$CH_3C\equiv CH$	**Propyne**	

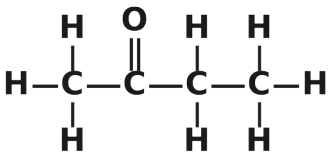
Worked examples: formula → name

Formula	How to think	Name
$CH_3–CH_2–Br$	2 C → ethane; Br → prefix bromo	**Bromoethane**
HCHO	1 C (the CHO carbon) → methane; –CHO → -al	**Methanal**
$HC\equiv C–CH_2–CH_2–CH_2–CH_3$	6 C → hexane; triple bond → -yne	**Hexyne**
$CH_3–CO–CH_2–CH_3$	4 C → butane; –CO– → -one	**Butanone**
$CH_3–CH_2–CH_2–CH_2–CH_2–CHO$	6 C (count the CHO carbon) → -al	**Hexanal**
$CH_3–CH_2–CH_2–CH_2–CH_2–Br$	5 C → pentane; Br → bromo	**Bromopentane**
$CH_3–CH_2–CH_2–CH_2–OH$	4 C → butane; –OH → -ol	**Butanol**
$CH_3–COOH$	2 C → ethane; –COOH → -oic acid	**Ethanoic acid**
$CH_3–CH=CH–CH_3$	4 C → butane; double bond → -ene	**Butene**

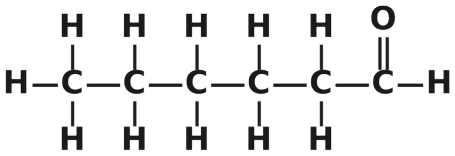
Worked examples: name → structure



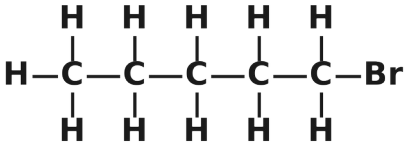
Ethanoic acid, CH_3COOH



Butanone, $CH_3COCH_2CH_3$

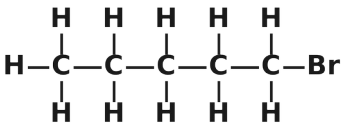


Hexanal, $CH_3(CH_2)_4CHO$

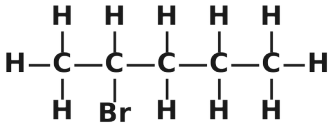


Bromopentane, $CH_3(CH_2)_4Br$

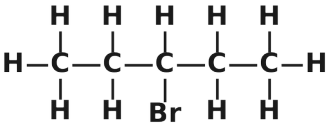
Are structural isomers possible for bromopentane? Yes. The Br atom can be attached to different carbon atoms, and the carbon chain can also be branched. For example:



Br on the 1st carbon



Br on the 2nd carbon



Br on the 3rd carbon

EXAM TIP

Position numbers (as in propan-1-ol or but-2-ene) are **not required in Class 10**. Follow NCERT and simply write propanol, butene, chloropropane, etc.

COMMON MISTAKE

- There is no "methanone" or "ethanone" — a ketone needs **at least 3 carbon atoms**.
- CH_3CHO is **ethanal** (2 C), not methanal; HCOOH is **methanoic acid** (1 C). Always count the carbon of –CHO and –COOH.
- Drop the "e" only before a vowel: propan**ol**, propan**al**, propan**one** — but **chloro**propane keeps the full name because "chloro" is a prefix.

10 CHEMICAL PROPERTIES OF CARBON COMPOUNDS

Most of the fuels we use are carbon or its compounds. We study four important types of reactions: **combustion, oxidation, addition** and **substitution**.

10.1 Combustion

DEFINITION

Combustion: The burning of a substance in oxygen (air) to give **heat and light**. It is an oxidation reaction and is exothermic.

Carbon, in **all its allotropic forms**, burns in oxygen to give carbon dioxide along with heat and light. Most carbon compounds also release a large amount of heat and light on burning.

$C + O_2 \rightarrow$	CO_2 + heat and light	*Carbon (any allotrope)*
$CH_4 + 2O_2 \rightarrow$	$CO_2 + 2H_2O$ + heat and light	*Methane (CNG, biogas)*
$C_2H_5OH + 3O_2 \rightarrow$	$2CO_2 + 3H_2O$ + heat and light	*Ethanol*
$2C_4H_{10} + 13O_2 \rightarrow$	$8CO_2 + 10H_2O$ + heat	*Butane (LPG)*

NCERT writes the methane and ethanol equations without coefficients and asks you to balance them — the balanced forms are given above.

Complete and incomplete combustion

Complete combustion	Incomplete combustion
Sufficient supply of oxygen (air)	Limited supply of oxygen
Products: CO_2 and H_2O	Products: carbon monoxide (CO, poisonous) and/or carbon (soot)
Clean **blue** flame; no smoke	**Yellow, sooty** (smoky) flame
More heat produced	Less heat produced; fuel is wasted
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$

Nature of the flame

- **Saturated hydrocarbons** generally burn with a **clean flame**.
- **Unsaturated** carbon compounds burn with a **yellow flame with lots of black smoke** — this leaves a sooty deposit on a metal plate held above the flame.
- Limiting the supply of air causes **incomplete combustion**, so even saturated hydrocarbons then give a sooty flame.

- A gas or kerosene stove has **inlets for air**, so that an oxygen-rich mixture burns with a clean **blue flame**. If the bottoms of cooking vessels are getting **blackened**, the air holes are blocked and fuel is being wasted.
- Coal and petroleum contain some **nitrogen and sulphur**. Their combustion forms **oxides of sulphur and nitrogen**, which are major pollutants of our environment.

> ACTIVITY **Burning carbon compounds (with your teacher's help)**
>
> - Burn **naphthalene**, **camphor** and **alcohol** one by one on a spatula and hold a metal plate above each flame. Naphthalene and camphor burn with a **yellow, sooty flame** and leave a **black deposit**; alcohol burns with a **clean flame** and leaves no deposit.
> - **Bunsen burner:** with the air hole **closed**, the flame is **yellow and sooty** (incomplete combustion); with the air hole **open**, it is **blue** (complete combustion).

Why do some substances burn with a flame and others without?

- A **flame** is produced only when **gaseous substances** burn.
- When wood or charcoal is ignited, the volatile substances present vaporise and burn with a flame in the beginning. Coal or charcoal in an **angithi** often just **glows red** and gives out heat without a flame.
- A **luminous (yellow) flame** is seen when the atoms of the gaseous substance are heated and start to glow. The yellow colour of a **candle flame** comes from glowing, unburnt **carbon (soot)** particles.
- The colour produced by each element is characteristic of that element — a copper wire heated in a gas flame gives a **bluish-green** colour.

Why are carbon and its compounds used as fuels?

- They burn in air releasing a **large amount of heat and light** (high calorific value).
- Their combustion can be **controlled**, and their ignition temperature is moderate.
- Many of them (wood, coal, LPG, petrol, CNG) are **easily available** and easy to store and transport.
- Saturated hydrocarbons burn with a clean flame, giving carbon dioxide and water.

Fossil fuels — formation of coal and petroleum

- **Coal** is the remains of trees, ferns and other plants that lived millions of years ago. They were crushed into the earth (perhaps by earthquakes or volcanic eruptions), pressed down by layers of earth and rock, and slowly decayed into coal.
- **Petroleum and natural gas** are the remains of millions of tiny plants and animals that lived in the sea. When they died, their bodies sank to the sea bed and were covered by silt. Bacteria attacked the remains and, under high pressure, turned them into oil and gas. The silt was slowly compressed into rock; the oil and gas seeped into its porous parts and were trapped like water in a sponge.
- They are called **fossil fuels** because they were formed from the remains (fossils) of living organisms over millions of years.

Oxy-acetylene welding

A mixture of **ethyne (acetylene) and oxygen** is burnt for welding. Complete combustion gives a very hot flame (above 3000 °C) that melts metals:

$$2C_2H_2 + 5O_2 \longrightarrow 4CO_2 + 2H_2O + \text{heat}$$

Ethyne burns in pure oxygen

Why not air? Air contains only about **21% oxygen** (it is mostly nitrogen). A mixture of ethyne and air burns **incompletely** with a sooty flame, and the temperature reached is not high enough for welding.

10.2 Oxidation

Combustion is **complete oxidation**. Carbon compounds can also be oxidised in a controlled way: **alcohols are converted into carboxylic acids** by oxidising agents.

$CH_3–CH_2OH \xrightarrow[\text{or acidified } K_2Cr_2O_7 + \text{heat}]{\text{Alkaline } KMnO_4 + \text{heat}} CH_3COOH$ *Ethanol → Ethanoic acid*

Writing the oxygen supplied by the oxidising agent as [O]: $CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O$

DEFINITION

Oxidising agents: Substances that are capable of **adding oxygen** to others (or removing hydrogen). Examples: **alkaline potassium permanganate** ($KMnO_4$) and **acidified potassium dichromate** ($K_2Cr_2O_7$).

ACTIVITY **Oxidation of ethanol**

Warm about 3 mL of ethanol in a water bath and add a 5% solution of **alkaline $KMnO_4$** drop by drop.

- At first, the **purple colour disappears** — $KMnO_4$ is used up in oxidising ethanol.
- When excess $KMnO_4$ is added, the **colour persists** — there is no ethanol left to react.

REMEMBER **Why is ethanol → ethanoic acid an oxidation?**

Ethanol (C_2H_6O) **gains one oxygen atom** and **loses two hydrogen atoms** to become ethanoic acid ($C_2H_4O_2$). Addition of oxygen / removal of hydrogen = **oxidation**.

10.3 Addition reaction

DEFINITION

Addition reaction: A reaction in which atoms or molecules **add across a double or triple bond**, converting an unsaturated compound into a saturated one. Only **unsaturated** compounds show addition reactions.

Unsaturated hydrocarbons add **hydrogen** in the presence of catalysts such as **palladium or nickel** to give saturated hydrocarbons. This addition of hydrogen is called **hydrogenation**.

$R_2C=CR_2 + H_2 \xrightarrow{\text{Nickel catalyst}} R_2CH–CHR_2$ *General reaction (NCERT)*

$CH_2=CH_2 + H_2 \xrightarrow{\text{Ni, heat}} CH_3–CH_3$ *Ethene → Ethane*

$HC\equiv CH + 2H_2 \xrightarrow{\text{Ni, heat}} CH_3–CH_3$ *Ethyne → Ethane*

DEFINITION

Catalyst: A substance that causes a reaction to occur or to proceed at a different rate **without the reaction itself affecting it** (it is not used up).

Industrial application: hydrogenation of vegetable oils

- Vegetable oils generally have **long unsaturated** carbon chains (liquids); animal fats have **saturated** carbon chains (solids).
- Vegetable oils are hydrogenated using a **nickel catalyst** to make **vanaspati ghee**.
- Animal fats generally contain **saturated** fatty acids, which are said to be harmful for health. Oils containing **unsaturated** fatty acids should be chosen for cooking.

Vegetable oil + H_2 $\xrightarrow{\text{Ni catalyst, heat}}$ Vanaspati ghee *(unsaturated, liquid) → (saturated, solid)*

EXAM TIP **Bromine water — a test based on addition**

Reddish-brown bromine adds across the double bond, so the colour disappears:

$$CH_2{=}CH_2 + Br_2 \longrightarrow CH_2Br{-}CH_2Br$$ *colourless product*

Cooking oil (unsaturated) decolourises bromine water; butter (saturated animal fat) does not. Among C_2H_6, C_3H_8, C_3H_6, C_2H_2 and CH_4, only **C_3H_6 and C_2H_2** (unsaturated) undergo addition reactions.

10.4 Substitution reaction

Saturated hydrocarbons are fairly unreactive and are inert in the presence of most reagents. However, in the presence of **sunlight**, **chlorine** reacts with hydrocarbons in a very fast reaction — chlorine **replaces the hydrogen atoms one by one**.

DEFINITION

Substitution reaction: A reaction in which **one type of atom or group of atoms takes the place of another**.

$$CH_4 + Cl_2 \xrightarrow{\text{sunlight}} CH_3Cl + HCl$$ *Chloromethane*

$$CH_3Cl + Cl_2 \xrightarrow{\text{sunlight}} CH_2Cl_2 + HCl$$ *Dichloromethane*

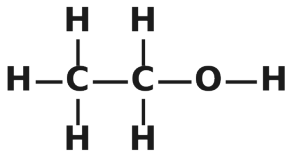
$$CH_2Cl_2 + Cl_2 \xrightarrow{\text{sunlight}} CHCl_3 + HCl$$ *Trichloromethane (chloroform)*

$$CHCl_3 + Cl_2 \xrightarrow{\text{sunlight}} CCl_4 + HCl$$ *Tetrachloromethane*

With higher homologues of alkanes, a number of products are usually formed. For example, ethane gives chloroethane first: $C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl$ (in sunlight).

Addition reaction	Substitution reaction
Shown by **unsaturated** compounds	Shown by **saturated** compounds
Atoms are added across C=C or C≡C; nothing is removed	One atom or group **replaces** another
Usually gives a single product	Often gives a mixture of products
Unsaturated compound becomes saturated	Compound remains saturated
Example: ethene + H_2 → ethane (Ni catalyst)	Example: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (sunlight)

11 ETHANOL



Structure of ethanol

Ethanol (ethyl alcohol), **C_2H_5OH** or **CH_3CH_2OH**, is one of the two commercially important carbon compounds studied in this chapter. It is commonly called **alcohol**.

Functional group: **–OH** (alcohol)

11.1 Physical properties

- Colourless **liquid** at room temperature with a characteristic smell and burning taste.
- Melting point **156 K** (−117 °C); boiling point **351 K** (78 °C) — it is volatile.

- **Soluble in water in all proportions.**
- **Neutral** — it has no effect on litmus.
- A good **solvent** for many substances; highly inflammable (burns with a blue flame).

11.2 Uses

- Active ingredient of all **alcoholic drinks**.
- Being a good solvent, it is used in medicines such as **tincture iodine, cough syrups** and many **tonics**.
- An important **industrial solvent**, and used to make other chemicals (ethanoic acid, esters).
- **Fuel:** added to petrol as a cleaner fuel — India now sells **E20 petrol** (20% ethanol + 80% petrol).
- **Antiseptic:** used to clean wounds and in hand sanitisers.

11.3 Chemical properties

(i) Combustion

$$C_2H_5OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O + \text{heat and light}$$

Burns with a clean blue flame

(ii) Reaction with sodium

$$2Na + 2CH_3CH_2OH \longrightarrow 2CH_3CH_2O^-Na^+ + H_2\uparrow$$

Sodium ethoxide + hydrogen

- Alcohols react with sodium to give **hydrogen gas**; with ethanol, the other product is **sodium ethoxide**.
- Test for hydrogen: bring a burning splinter near the gas — it burns with a **pop sound**.
- Acids also give hydrogen with metals. (Teacher demonstration only — sodium reacts vigorously.)

(iii) Dehydration — formation of an unsaturated hydrocarbon

$$CH_3{-}CH_2OH \xrightarrow[443\ K]{\text{Hot conc. } H_2SO_4} CH_2{=}CH_2 + H_2O$$

Ethanol → Ethene

Heating ethanol at **443 K** with **excess concentrated sulphuric acid** removes a water molecule from it. Concentrated H_2SO_4 acts as a **dehydrating agent**.

(iv) Oxidation and (v) esterification

Ethanol is oxidised to ethanoic acid by alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ (Section 10.2), and reacts with ethanoic acid to form a sweet-smelling ester (Section 12).

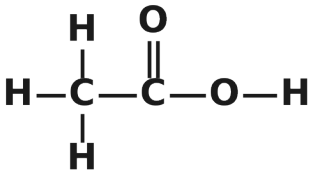
HEALTH ALERT **How do alcohols affect living beings?**

- Large quantities of ethanol **slow metabolic processes** and **depress the central nervous system** → lack of coordination, mental confusion, drowsiness, lowering of normal inhibitions and finally stupour. The person may feel relaxed but does not realise that judgement, sense of timing and muscular coordination are seriously impaired.
- Intake of even a small quantity of pure ethanol (**absolute alcohol**) can be lethal, and long-term consumption leads to many health problems.
- **Methanol** is deadly even in very small quantities. In the liver it is oxidised to **methanal**, which reacts rapidly with the components of cells and coagulates the protoplasm (like an egg on cooking). Methanol also affects the **optic nerve**, causing **blindness**.
- **Denatured alcohol:** industrial ethanol is made unfit for drinking by adding poisonous substances like **methanol**; a **blue dye** is added so that it can be identified easily.

DO YOU KNOW? **Alcohol as a fuel**

Sugarcane plants are among the most efficient converters of sunlight into chemical energy. Sugarcane juice is used to make **molasses**, which is **fermented** to give ethanol. Ethanol is a **cleaner fuel** — in sufficient oxygen it gives only carbon dioxide and water on burning — so many countries, including India, add it to petrol.

12 ETHANOIC ACID



Structure of ethanoic acid

Ethanoic acid (acetic acid), **CH_3COOH**, belongs to the group of organic acids called **carboxylic acids** (functional group **–COOH**).

A **5–8% solution** of acetic acid in water is called **vinegar**; it is widely used as a **preservative in pickles**.

12.1 Physical properties

- Colourless liquid with a **sour taste** and a pungent, **vinegar-like smell**.
- Melting point **290 K** (17 °C), so pure ethanoic acid often **freezes during winter** in cold climates — hence the name **glacial acetic acid**. Boiling point 391 K (118 °C).
- Mixes with water in all proportions.
- **Weak acid:** carboxylic acids are only partially ionised, unlike mineral acids such as HCl, which are completely ionised.

ACTIVITY **Weak acid vs strong acid**

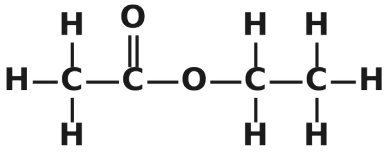
Test dilute acetic acid and dilute hydrochloric acid with **blue litmus** and **universal indicator**. Both turn blue litmus red, but universal indicator shows a **lower pH for HCl** — so ethanoic acid is the **weaker** acid.

12.2 Chemical properties

(i) Esterification

$$CH_3COOH + CH_3CH_2OH \xrightarrow[\text{warm}]{\text{Acid (conc. } H_2SO_4\text{)}} CH_3COOCH_2CH_3 + H_2O$$

Ethyl ethanoate (an ester)



Ethyl ethanoate (ethyl acetate)

- Esters are formed by the reaction of an **acid and an alcohol** in the presence of an acid catalyst.
- Esters are **sweet (fruity)-smelling** substances, used in making **perfumes** and as **flavouring agents**.

ACTIVITY **Making an ester**

Take 1 mL absolute ethanol and 1 mL glacial acetic acid with a few drops of concentrated H_2SO_4 in a test tube. Warm in a **water bath** for at least five minutes, then pour into a beaker containing 20–50 mL of water and smell — a **sweet, fruity smell** shows that an ester has formed.

(ii) Saponification

$$CH_3COOC_2H_5 \xrightarrow{NaOH} C_2H_5OH + CH_3COONa$$

Ester → alcohol + sodium salt

On treating with sodium hydroxide (an alkali), the ester is converted back to alcohol and the sodium salt of the carboxylic acid. This reaction is called **saponification** because it is used in the **preparation of soap**.

(iii) Reaction with a base (neutralisation)

$$NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$$

Sodium ethanoate (sodium acetate)

(iv) Reaction with carbonates and hydrogencarbonates

$$2CH_3COOH + Na_2CO_3 \longrightarrow 2CH_3COONa + H_2O + CO_2\uparrow$$

Brisk effervescence

$$CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2\uparrow$$

Brisk effervescence

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3\downarrow + H_2O$$

Lime water turns milky

Ethanoic acid reacts with carbonates and hydrogencarbonates to give a **salt, carbon dioxide and water**. The CO_2 turns lime water milky.

> **DO YOU KNOW?**
> Like other acids, ethanoic acid also reacts with active metals to give hydrogen: $2CH_3COOH + 2Na \rightarrow 2CH_3COONa + H_2\uparrow$. So sodium metal cannot be used to tell ethanol and ethanoic acid apart — both give hydrogen!

12.3 Uses

- As **vinegar** — preservative in pickles and flavouring in food.
- To make **esters** used in perfumes and flavourings.
- In the manufacture of plastics, dyes, synthetic fibres and medicines; as a laboratory reagent.

12.4 How to distinguish ethanol and ethanoic acid

Test / property	Ethanol	Ethanoic acid
Smell	Characteristic, pleasant smell	Pungent, vinegar-like smell
Melting point	156 K — stays liquid in winter	290 K — may freeze in cold winters
Boiling point	351 K	391 K
Litmus	No change (neutral)	Turns blue litmus **red** (acidic)
Sodium hydrogencarbonate ($NaHCO_3$)	No reaction	**Brisk effervescence** of CO_2; lime water turns milky
Sodium metal	H_2 gas evolved	H_2 gas evolved (not useful to distinguish)

> **EXAM TIP**
> For "How would you distinguish experimentally between an alcohol and a carboxylic acid?" — the best answer is the **$NaHCO_3$ test**: the acid gives brisk effervescence of CO_2 (lime water turns milky); the alcohol gives none. Write the equation for full marks.

13 SOAPS AND DETERGENTS

Most dirt is **oily** in nature, and oil does not dissolve in water. So water alone cannot remove oily dirt — we need soaps or detergents.

ACTIVITY **Effect of soap on oil and water**

Take 10 mL water in each of two test tubes, A and B, and add a drop of cooking oil to both. Add a few drops of soap solution to B only and shake both equally. When left undisturbed, the oil layer separates **first in A** (no soap). In B, soap keeps the oil mixed with the water for longer — soap helps oil and water mix (forms an **emulsion**).

13.1 Soaps

DEFINITION

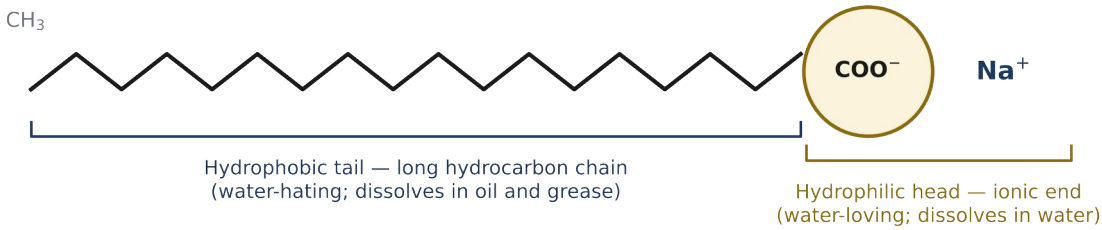
Soaps: Sodium or potassium salts of long-chain carboxylic acids (fatty acids). Example: sodium stearate, $C_{17}H_{35}COONa$.

Soap is made by heating oils or fats with sodium hydroxide — the saponification reaction:

$$\text{Oil or fat} + NaOH \xrightarrow{\text{heat}} \text{Soap} + \text{Glycerol}$$

Saponification

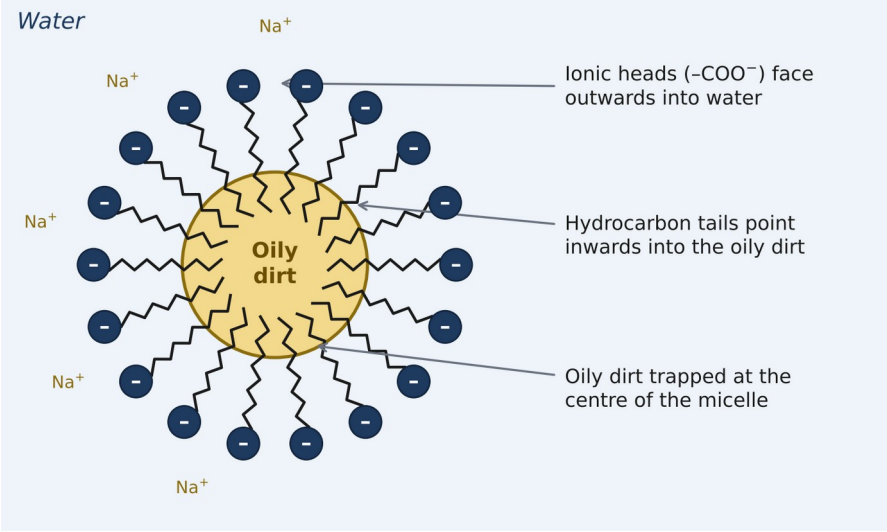
Sodium stearate: $CH_3\text{–}(CH_2)_{16}\text{–}COO^-Na^+$ ($C_{17}H_{35}COONa$)



A soap molecule has two ends with different properties

- **Ionic end ($-COO^-Na^+$)** — **hydrophilic** (water-loving): it interacts with water.
- **Long hydrocarbon chain** — **hydrophobic** (water-hating): it interacts with oil and grease (hydrocarbons).

13.2 Micelles and the cleansing action of soap

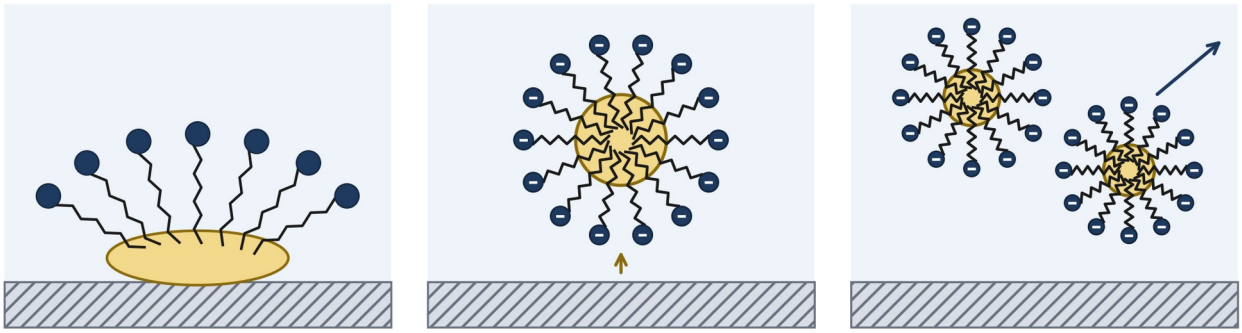


A soap micelle with oily dirt trapped at its centre

- At the **surface** of water, soap molecules align with the ionic end in the water and the hydrocarbon tail sticking out of the water.
- **Inside** water, soap molecules form clusters in which the hydrophobic tails are in the **interior** and the ionic ends are on the **surface**. Such a cluster is called a **micelle**.
- The **oily dirt** collects in the **centre** of the micelle, and the micelles form an **emulsion** in water.

- Micelles stay in solution as a **colloid** and do not come together to precipitate, because of **ion–ion repulsion** (their surfaces carry similar negative charges).
- The dirt held in the micelles is therefore easily **rinsed away**.
- Soap micelles are large enough to **scatter light** — that is why a soap solution looks **cloudy**.

Cleansing action — step by step



1. Soap tails stick to the oily dirt on the cloth
2. Agitation lifts the dirt; a micelle forms around it
3. Rinsing washes the micelles away — clean cloth

1. Soap dissolves in water; the **hydrocarbon tails** attach themselves to the **oily dirt**, while the ionic heads stay in water.
2. **Micelles** form with the oily dirt trapped at their centre — an emulsion of oil in water.
3. **Agitation** (rubbing, beating on a stone, scrubbing or a washing machine) loosens the dirt from the cloth fibres so the micelles can carry it away.
4. **Rinsing** with water washes away the micelles along with the dirt, leaving the cloth clean.

> REMEMBER **Three favourite questions**
>
> - **Why is agitation necessary?** It helps the soap micelles to dislodge the oily dirt from the cloth fibres and carry it into the water.
> - **Will micelles form in ethanol? No.** The hydrocarbon tail of soap is also soluble in ethanol, so soap dissolves as separate molecules and does not need to form clusters.
> - **Soap and litmus:** soap solution is **basic** — it turns **red litmus blue**; blue litmus stays blue. (Soap is the salt of a strong base, NaOH, and a weak fatty acid.)

13.3 Hard water and scum

- **Hard water** contains salts of **calcium and magnesium** (their hydrogencarbonates, chlorides and sulphates).
- Soap reacts with Ca^{2+} and Mg^{2+} ions to form an **insoluble** substance called **scum** (white, curdy precipitate). Foam forms with difficulty, so a **larger amount of soap** is needed, and the scum sticks to clothes.

$$2C_{17}H_{35}COONa + CaCl_2 \longrightarrow (C_{17}H_{35}COO)_2Ca\downarrow + 2NaCl$$ *Scum (calcium salt)*

$$2C_{17}H_{35}COONa + MgSO_4 \longrightarrow (C_{17}H_{35}COO)_2Mg\downarrow + Na_2SO_4$$ *Scum (magnesium salt)*

ACTIVITY **Soap and detergent in hard water**

- Shake a few drops of soap solution with **distilled (or rain) water** in one test tube and with **hard water** (from a tube well or hand pump) in another. Distilled water gives **more foam**; hard water gives a **white curdy precipitate**.
- Shake hard water with soap solution in one test tube and with detergent solution in another. The **detergent** gives plenty of foam and **no curdy solid**; the soap forms scum.
- If hard water is not available, it can be prepared by dissolving hydrogencarbonates, sulphates or chlorides of calcium or magnesium in water.

13.4 Detergents

- Detergents are generally **sodium salts of sulphonic acids** or **ammonium salts with chloride or bromide ions**. Both types have a **long hydrocarbon chain**.
- Their charged ends **do not form insoluble precipitates** with the calcium and magnesium ions of hard water, so they remain **effective in hard water**.
- Detergents are used to make **shampoos** and products for **cleaning clothes**.
- **Can a detergent tell us whether water is hard?** No — detergents lather well in both hard and soft water, so there is no difference to observe. Use **soap** instead: hard water gives little foam and forms scum.

Soaps	Detergents
Sodium or potassium salts of long-chain **carboxylic acids**	Sodium salts of long-chain **sulphonic acids**, or ammonium salts with chloride/bromide ions
Ionic group: $-COO^-Na^+$	Ionic group: $-SO_3^-Na^+$ (or an ammonium group)
Form **scum** with hard water; not effective in hard water	Do **not** form scum; effective even in hard water
Made from vegetable oils and animal fats	Made synthetically, mainly from petroleum products
Biodegradable	Some (branched-chain) detergents are not biodegradable and cause water pollution

14 NCERT ACTIVITIES AT A GLANCE

Activity	What we do	What we observe	Conclusion
Carbon around us	List 10 things used or eaten since morning	Most of them (food, clothes, fuel, paper) contain carbon	Carbon compounds are everywhere
Homologous series	Find the difference between CH_3OH, C_2H_5OH, C_3H_7OH, C_4H_9OH	Each differs by $-CH_2-$ and 14 u	They form a homologous series
Burning compounds	Burn naphthalene, camphor, alcohol; hold a plate above	Naphthalene, camphor: sooty flame, black deposit; alcohol: clean flame	Unsaturated / carbon-rich compounds burn with a sooty flame
Bunsen flame	Close and open the air hole	Closed: yellow sooty flame; open: blue flame	Enough air is needed for complete combustion
Oxidation of ethanol	Add alkaline $KMnO_4$ to warm ethanol	Purple colour disappears; persists when in excess	Ethanol → ethanoic acid; $KMnO_4$ is an oxidising agent
Sodium + ethanol	Drop a tiny piece of Na into ethanol (teacher)	Effervescence; gas burns with a pop	H_2 evolved; sodium ethoxide formed
Acid strength	Test dilute CH_3COOH and HCl with litmus and universal indicator	Both turn blue litmus red; HCl shows lower pH	Ethanoic acid is a weak acid
Esterification	Warm ethanol + acetic acid + conc. H_2SO_4	Sweet, fruity smell	An ester is formed

Activity	What we do	What we observe	Conclusion
Acid + carbonate	Add ethanoic acid to Na_2CO_3 / $NaHCO_3$	Brisk effervescence; lime water turns milky	CO_2 is evolved
Soap and oil	Shake oil + water with and without soap	Oil separates first without soap	Soap emulsifies oil
Soap in hard water	Soap with distilled water and with hard water	Less foam and curdy precipitate in hard water	Hard water forms scum with soap
Soap vs detergent	Soap and detergent with hard water	Detergent foams well, no scum	Detergents work in hard water

15 QUICK REVISION AND MIND MAP

15.1 Key definitions

Term	Meaning
Covalent bond	Bond formed by sharing a pair of electrons between two atoms
Allotropes	Different physical forms of the same element with similar chemical properties
Catenation	Ability of carbon to form bonds with other carbon atoms → chains, branches, rings
Tetravalency	Carbon has valency 4 — it can bond with four other atoms
Hydrocarbon	Compound of carbon and hydrogen only
Saturated compound	Only single bonds between carbon atoms (alkanes)
Unsaturated compound	Double or triple bond(s) between carbon atoms (alkenes, alkynes)
Structural isomers	Same molecular formula, different structures
Heteroatom	Atom other than C and H that replaces hydrogen in a carbon chain
Functional group	Atom/group that decides the characteristic properties of a carbon compound
Homologous series	Compounds with the same functional group; successive members differ by $-CH_2-$ (14 u)
Combustion	Burning in oxygen giving heat and light
Oxidising agent	Substance that adds oxygen to others (alkaline $KMnO_4$, acidified $K_2Cr_2O_7$)
Catalyst	Changes the rate of a reaction without itself being used up
Hydrogenation	Addition of hydrogen to unsaturated compounds (Ni/Pd catalyst)
Substitution	One atom/group takes the place of another ($CH_4 + Cl_2$ in sunlight)
Esterification	Acid + alcohol → ester + water (acid catalyst)
Saponification	Ester + NaOH → alcohol + sodium salt of acid; used to make soap
Soap	Sodium/potassium salt of a long-chain carboxylic acid
Detergent	Sodium salt of a long-chain sulphonic acid (or ammonium salt); works in hard water
Micelle	Cluster of soap molecules: tails inside (oil), ionic heads outside (water)
Scum	Insoluble precipitate formed by soap with Ca^{2+}/Mg^{2+} ions of hard water
Denatured alcohol	Industrial ethanol made unfit to drink with methanol and a blue dye
Glacial acetic acid	Pure ethanoic acid (freezes at 290 K in cold winters)
Vinegar	5–8% solution of acetic acid in water
Absolute alcohol	Pure ethanol

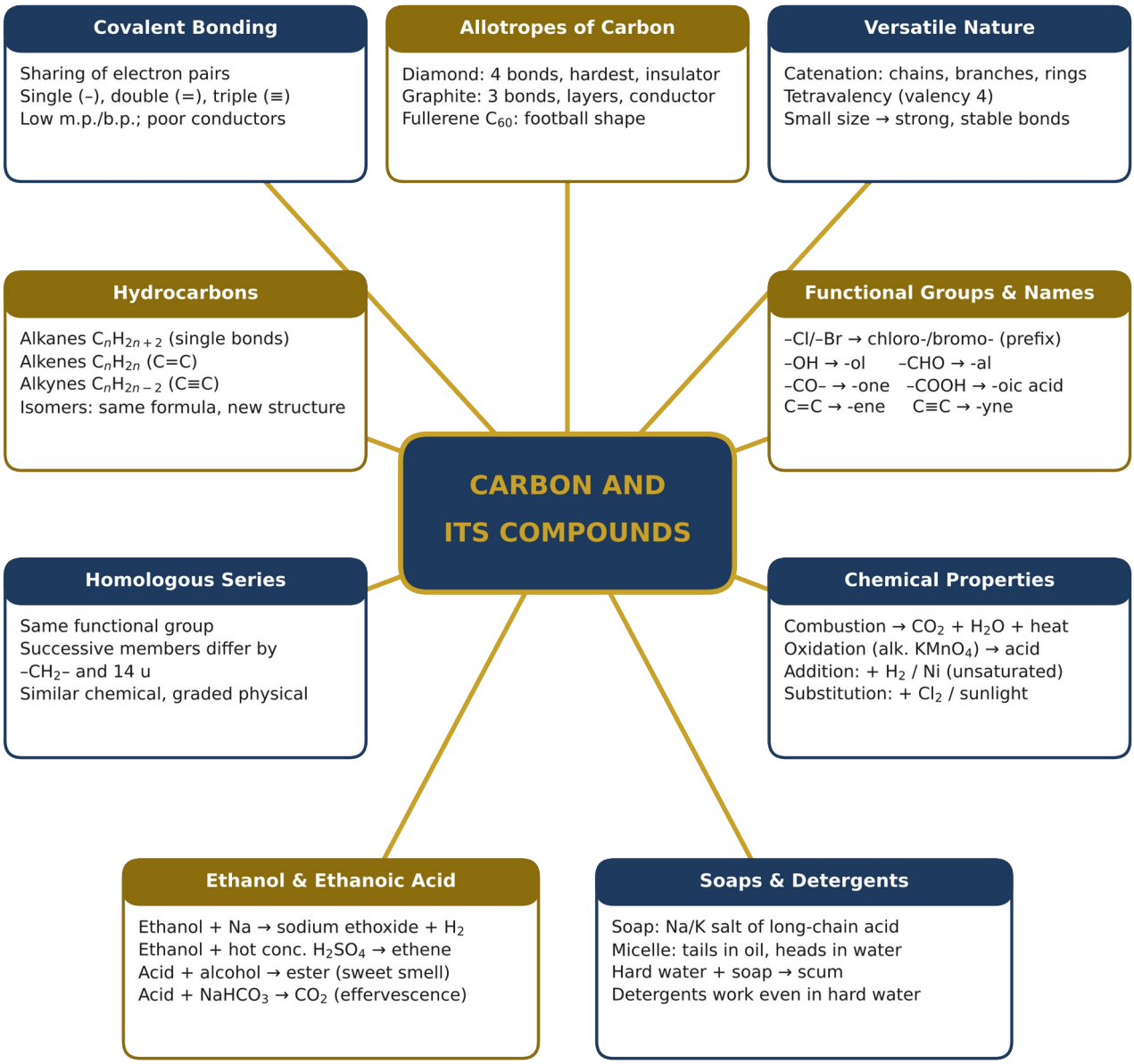
15.2 Reaction bank

#	Reaction	Equation	Condition
1	Combustion of methane	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat + light	Burning in air
2	Combustion of ethanol	$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ + heat + light	Burning in air
3	Oxidation of ethanol	$CH_3CH_2OH \rightarrow CH_3COOH$	Alk. $KMnO_4$ or acid. $K_2Cr_2O_7$, heat
4	Hydrogenation	$CH_2{=}CH_2 + H_2 \rightarrow CH_3{-}CH_3$	Ni or Pd catalyst
5	Bromine water test	$CH_2{=}CH_2 + Br_2 \rightarrow CH_2Br{-}CH_2Br$	Room temperature
6	Chlorination of methane	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$	Sunlight
7	Ethanol + sodium	$2Na + 2CH_3CH_2OH \rightarrow 2CH_3CH_2ONa + H_2$	—
8	Dehydration of ethanol	$CH_3CH_2OH \rightarrow CH_2{=}CH_2 + H_2O$	Hot conc. H_2SO_4, 443 K
9	Esterification	$CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$	Acid catalyst, warm
10	Saponification	$CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$	—
11	Neutralisation	$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$	—
12	Acid + carbonate	$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$	—
13	Acid + hydrogencarbonate	$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$	—
14	Scum formation	$2C_{17}H_{35}COONa + CaCl_2 \rightarrow (C_{17}H_{35}COO)_2Ca + 2NaCl$	Hard water

15.3 Numbers to remember

Fact	Value	Fact	Value
Carbon in earth's crust	0.02%	CO_2 in atmosphere	0.03%
Known carbon compounds (NCERT)	≈ 3 million	Silicon chain length	7–8 atoms
Mass difference in a homologous series	14 u	Wöhler made urea	1828
Melting point of ethanol	156 K	Boiling point of ethanol	351 K
Melting point of ethanoic acid	290 K	Boiling point of ethanoic acid	391 K
Vinegar	5–8% acetic acid	Dehydration of ethanol	443 K
Atoms in buckminsterfullerene	60 (C_{60})	Oxygen in air	≈ 21%

15.4 Chapter at a glance — mind map



16 NCERT QUESTIONS — SOLVED

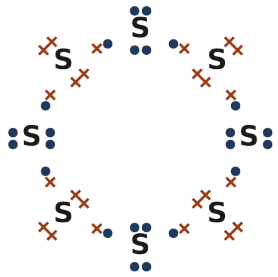
16.1 In-text questions

Q. What would be the electron dot structure of carbon dioxide, which has the formula CO_2?

Ans. Carbon shares two electrons with each oxygen atom, forming **two double bonds** (O=C=O). See the CO_2 structure in Section 2.4.

Q. What would be the electron dot structure of a molecule of sulphur, which is made up of eight atoms of sulphur? (Hint: the eight atoms are joined in a ring.)

Ans. Each S atom (2, 8, 6) forms single bonds with its two neighbours in the ring and keeps **two lone pairs**:



Electron dot structure of S_8 (crown-shaped ring drawn flat)

Q. How many structural isomers can you draw for pentane?

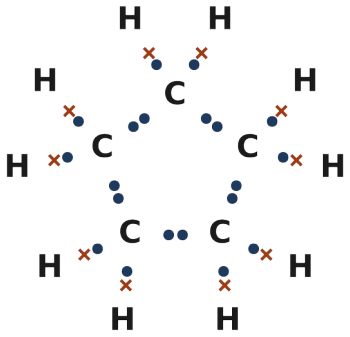
Ans. **Three:** n-pentane, isopentane (2-methylbutane) and neopentane (2,2-dimethylpropane). See Section 6.2.

Q. What are the two properties of carbon which lead to the huge number of carbon compounds we see around us?

Ans. (i) **Catenation** — carbon bonds with other carbon atoms to form long chains, branched chains and rings. (ii) **Tetravalency** — carbon can bond with four other atoms of carbon or other elements.

Q. What will be the formula and electron dot structure of cyclopentane?

Ans. Formula: C_5H_{10}. Five carbon atoms form a ring; each carbon is bonded to two other carbons and two hydrogens.



Electron dot structure of cyclopentane, C_5H_{10}

Q. Draw the structures of: (i) ethanoic acid (ii) bromopentane (iii) butanone (iv) hexanal. Are structural isomers possible for bromopentane?

Ans. The structures are drawn in Section 9 (worked examples: name → structure). **Yes**, bromopentane has structural isomers — the Br atom can be on the 1st, 2nd or 3rd carbon, and the chain can also be branched.

Q. How would you name: (i) $CH_3–CH_2–Br$ (ii) HCHO (iii) $HC{\equiv}C–CH_2–CH_2–CH_2–CH_3$?

Ans. (i) **Bromoethane** (ii) **Methanal** (iii) **Hexyne**

Q. Why is the conversion of ethanol to ethanoic acid an oxidation reaction?

Ans. Because oxygen is **added** (and hydrogen removed): $CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O$, using alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$.

Q. A mixture of oxygen and ethyne is burnt for welding. Can you tell why a mixture of ethyne and air is not used?

Ans. Air has only about 21% oxygen, so ethyne burns **incompletely** with a sooty flame and does not produce enough heat. Pure oxygen gives **complete combustion** and a very high temperature, which is needed to melt metals.

Q. How would you distinguish experimentally between an alcohol and a carboxylic acid?

Ans. Add **sodium hydrogencarbonate**. A carboxylic acid gives brisk effervescence of CO_2, which turns lime water milky ($CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$); an alcohol shows no reaction. (Litmus also helps: the acid turns blue litmus red.)

Q. What are oxidising agents?

Ans. Substances that **add oxygen** to other substances, e.g. alkaline $KMnO_4$ and acidified $K_2Cr_2O_7$, which oxidise alcohols to acids.

Q. Would you be able to check if water is hard by using a detergent?

Ans. No. Detergents do not form scum with Ca^{2+}/Mg^{2+} ions and lather well in both hard and soft water, so no difference is seen.

Q. Why is agitation necessary to get clean clothes?

Ans. Agitation (beating, scrubbing, washing machine) helps the soap micelles to **loosen and lift the oily dirt** from the cloth fibres so that it can be washed away.

16.2 Exercise questions

Q. 1. Ethane, with the molecular formula C_2H_6, has (a) 6 (b) 7 (c) 8 (d) 9 covalent bonds.

Ans. (b) 7 — one C–C bond and six C–H bonds.

Q. 2. Butanone is a four-carbon compound with the functional group (a) carboxylic acid (b) aldehyde (c) ketone (d) alcohol.

Ans. (c) Ketone

Q. 3. While cooking, if the bottom of the vessel is getting blackened on the outside, it means that (a) the food is not cooked completely (b) the fuel is not burning completely (c) the fuel is wet (d) the fuel is burning completely.

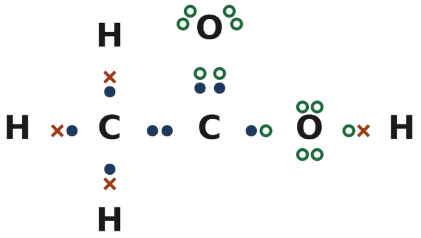
Ans. (b) The fuel is not burning completely (the air holes are blocked).

Q. 4. Explain the nature of the covalent bond using the bond formation in CH_3Cl.

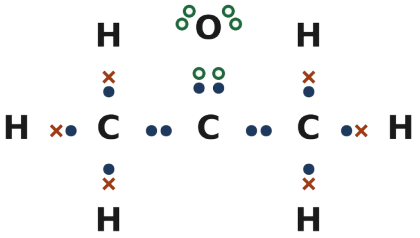
Ans. Carbon (2, 4) needs 4 electrons and cannot gain or lose them easily. In CH_3Cl, carbon **shares** one electron with each of three H atoms and one with a Cl atom, forming **four single covalent bonds**. Each shared pair belongs to both atoms, so C and Cl complete their octets and each H completes its duplet. No ions are formed — hence CH_3Cl is a covalent compound with a low boiling point that does not conduct electricity. (Electron dot structure: Section 2.4.)

Q. 5. Draw the electron dot structures for (a) ethanoic acid (b) H_2S (c) propanone (d) F_2.

Ans. H_2S and F_2 are drawn in Section 2.4. Ethanoic acid and propanone are shown below (• = electron of C, × = electron of H, ○ = electron of O):



(a) Ethanoic acid, CH_3COOH



(c) Propanone, CH_3COCH_3

Q. 6. What is a homologous series? Explain with an example.

Ans. A series of compounds in which the same functional group substitutes for hydrogen in a carbon chain; successive members differ by $-CH_2-$ (14 u), have similar chemical properties and show a gradation in physical properties. Example: alkanes — CH_4, C_2H_6, C_3H_8, C_4H_{10}; or alcohols — CH_3OH, C_2H_5OH, C_3H_7OH.

Q. 7. How can ethanol and ethanoic acid be differentiated on the basis of their physical and chemical properties?

Ans. Physical: ethanol has a pleasant smell, melting point 156 K, boiling point 351 K; ethanoic acid has a vinegar smell, melting point 290 K (freezes in cold winters), boiling point 391 K. Chemical: ethanoic acid turns blue litmus red and gives brisk effervescence with $NaHCO_3$; ethanol does neither. (See the table in Section 12.4.)

Q. 8. Why does micelle formation take place when soap is added to water? Will a micelle be formed in other solvents such as ethanol also?

Ans. A soap molecule has a hydrophobic hydrocarbon tail and a hydrophilic ionic head. In water, the tails avoid water and cluster together inside while the ionic heads face the water — forming micelles. **No**, micelles are not formed in ethanol, because the hydrocarbon tail is also soluble in ethanol, so soap dissolves as separate molecules.

Q. 9. Why are carbon and its compounds used as fuels for most applications?

Ans. They burn in air releasing a large amount of heat and light, their combustion can be controlled, they have a suitable ignition temperature, and many are easily available, cheap and easy to store and transport.

Q. 10. Explain the formation of scum when hard water is treated with soap.

Ans. Hard water contains Ca^{2+} and Mg^{2+} ions. These react with soap to form insoluble calcium and magnesium salts of fatty acids, which separate as a curdy white precipitate called scum: $2C_{17}H_{35}COONa + CaCl_2 \rightarrow (C_{17}H_{35}COO)_2Ca\downarrow + 2NaCl$.

Q. 11. What change will you observe if you test soap with litmus paper (red and blue)?

Ans. Soap solution is basic: **red litmus turns blue**; blue litmus shows no change.

Q. 12. What is hydrogenation? What is its industrial application?

Ans. Hydrogenation is the addition of hydrogen to an unsaturated compound in the presence of a catalyst such as nickel or palladium. Industrially, vegetable oils are hydrogenated with a nickel catalyst to make **vanaspati ghee**.

Q. 13. Which of the following hydrocarbons undergo addition reactions: C_2H_6, C_3H_8, C_3H_6, C_2H_2 and CH_4?

Ans. **C_3H_6 (propene)** and **C_2H_2 (ethyne)** — they are unsaturated (C_nH_{2n} and C_nH_{2n-2}).

Q. 14. Give a test that can be used to differentiate chemically between butter and cooking oil.

Ans. Add bromine water (or alkaline $KMnO_4$) to each. **Cooking oil** (unsaturated) **decolourises** it; **butter** (saturated) does not.

Q. 15. Explain the mechanism of the cleaning action of soaps.

Ans. The hydrophobic tails of soap molecules dissolve in the oily dirt while the ionic heads stay in water. Micelles form with the dirt trapped at the centre, giving an emulsion. Agitation detaches the dirt from the cloth, and the micelles (which repel one another and stay suspended) are washed away on rinsing. (See Section 13.2.)

17 PRACTICE ZONE WITH ANSWER KEY

A. Multiple choice questions

1. The number of covalent bonds in pentane (C_5H_{12}) is
   (a) 5 **(b)** 12 **(c)** 16 **(d)** 17
2. Which of the following is an unsaturated hydrocarbon?
   (a) C_3H_8 **(b)** C_4H_{10} **(c)** C_3H_4 **(d)** C_5H_{12}
3. The IUPAC name of CH_3COCH_3 is
   (a) propanal **(b)** propanone **(c)** propanol **(d)** propanoic acid
4. The general formula of alkynes is
   (a) C_nH_{2n+2} **(b)** C_nH_{2n} **(c)** C_nH_{2n-2} **(d)** C_nH_n
5. The next higher homologue of C_2H_5OH is
   (a) CH_3OH **(b)** C_3H_7OH **(c)** C_4H_9OH **(d)** C_2H_5CHO
6. Vinegar is a solution of
   (a) 50–60% acetic acid in alcohol **(b)** 5–8% acetic acid in water
   (c) 5–8% acetic acid in alcohol **(d)** 50–60% acetic acid in water
7. Which compound will decolourise bromine water?
   (a) CH_4 **(b)** C_2H_6 **(c)** C_2H_4 **(d)** C_3H_8
8. Conversion of vegetable oil into vanaspati ghee is an example of
   (a) substitution **(b)** addition (hydrogenation)
   (c) oxidation **(d)** saponification
9. Graphite conducts electricity because
   (a) it is a metal **(b)** it has free (mobile) electrons
   (c) its layers are held by strong forces **(d)** it contains ions
10. When ethanoic acid reacts with sodium hydrogencarbonate, the gas evolved is
    (a) H_2 **(b)** O_2 **(c)** CO_2 **(d)** CO
11. Heating ethanol with excess conc. H_2SO_4 at 443 K gives
    (a) ethane **(b)** ethene **(c)** ethyne **(d)** ethanal
12. Soap forms scum in hard water because of
    (a) Na^+ and K^+ ions **(b)** Ca^{2+} and Mg^{2+} ions **(c)** Cl^- and SO_4^{2-} ions **(d)** H^+ ions

B. Assertion–Reason questions

Choose: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. **Assertion (A):** Carbon forms covalent compounds. **Reason (R):** Carbon has four valence electrons and cannot easily gain or lose four electrons.
2. **Assertion (A):** Diamond does not conduct electricity, but graphite does. **Reason (R):** In graphite each carbon is bonded to only three others, leaving free electrons; in diamond all four valence electrons are used in bonding.
3. **Assertion (A):** Ethene undergoes addition reactions. **Reason (R):** Ethene is a saturated hydrocarbon.
4. **Assertion (A):** Soap is not effective in hard water. **Reason (R):** Soap reacts with calcium and magnesium ions to form insoluble scum.
5. **Assertion (A):** Butane and isobutane have the same molecular formula. **Reason (R):** Butane and isobutane are successive members of a homologous series.
6. **Assertion (A):** Esters are used in making perfumes. **Reason (R):** Esters are sweet-smelling substances.

C. Case-based question

CASE STUDY **Read and answer**

Riya noticed that when the flame of her family's gas stove turned yellow, the bottom of the cooking pot became black and the gas cylinder seemed to finish sooner. Her teacher explained that the burner's air holes were clogged.

1. What type of combustion was taking place when the flame was yellow?
2. Name the poisonous gas that can form in this situation.
3. Write the balanced equation for the complete combustion of methane.
4. Suggest one way to get back a clean blue flame and save fuel.

Answer key

Section	Answers
A. MCQs	1 (c) 16 2 (c) C_3H_4 3 (b) 4 (c) 5 (b) 6 (b) 7 (c) 8 (b) 9 (b) 10 (c) 11 (b) 12 (b)
B. Assertion–Reason	1 (a) 2 (a) 3 (c) 4 (a) 5 (c) — they are isomers, not homologues 6 (a)
C. Case study	1. Incomplete combustion 2. Carbon monoxide (CO) 3. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat and light 4. Clean the air holes so that enough air (oxygen) mixes with the gas.

— All the best! —